



Normalization

Normalisation is a process which we analyze and alter a database relation in order to get more concise and organized data structures. Normalised data is stable and has a natural structure. We call a relation normalized if:

- it does not contain any redundancy
- it does not cause maintenance problems
- it is an accurate representation of the data

Relations that aren't normalised contain non-atomic attributes and therefore can contain redundant information. Detailed planning of the ERM can help creating normalised relations. The following steps will explain how existing relations can be normalised step by step.

Dependencies

In order to be able to normalise a relation according to the three normal forms, we must first understand the concept of dependency between attributes within a relation.

Functional dependency:

If A and B are attributes of relation R, B is functionally dependent on A (denoted $A \twoheadrightarrow B$), if each value of A in R is associated with exactly one value of B in R.

Example:

ID	Name
S1	Meier
S2	Weber

The attribute Name is functionally dependent of attribute ID ($ID \twoheadrightarrow Name$).

Identification key:

If every attribute B of R is functionally dependent of A, then attribute A is a primary key.



Example:

ID	Name	Surname
S1	Meier	Hans
S2	Weber	Ueli

Attribute ID is the identification key

Full functional dependency:

We talk about full functional dependency if attribute B is functional dependent on A, if A is a composite primary key and B is not already functional dependent on parts of A.

Example:

IDStudent	Name	IDProfessor	Grade
S1	Meier	P2	5
S2	Weber	P1	6

The attribute Grade is fully functional dependent on the attributes IDStudent and IDProfessor.

Transitive dependency:

If A determines B and B determines C then C is determined by (dependent on) A. We write $A \rightarrow B$ and $B \rightarrow C$ but not $B \rightarrow A$.

Example:

ID	Name	Konto_Nr	Bank_Code_No	Bank
L1	Meier	1234-5	836	UBS
L2	Weber	5432-1	835	CS



There is a transitive dependency between Bank_Code_No and Bank because Bank_Code_No is not the primary key of the relation.

First normal form (1NF)

First normal form:

A relation is in first normal form if every attribute in every row can contain only one single (atomic) value. A university uses the following relation:

Student (Surname, Name, Skills)

The attribute Skills can contain multiple values and therefore the relation is not in the first normal form.

But the attributes Name and Surname are atomic attributes that can contain only one value.

Students

FirstName	LastName	Knowledge
Thomas	Mueller	Java, C++, PHP
Ursula	Meier	PHP, Java
Igor	Mueller	C++, Java

Startsituation

Result after Normalisation

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Students

FirstName	LastName	Knowledge
Thomas	Mueller	C++
Thomas	Mueller	PHP
Thomas	Mueller	Java
Ursula	Meier	Java
Ursula	Meier	PHP
Igor	Mueller	Java
Igor	Mueller	C++

Example First normal form

To get to the first normal form (1NF) we must create a separate tuple for each value of the multivalued attribute



Second normal form (2NF)

Second normal form:

A relation is in second normal form if it is in 1NF and every non key attribute is fully functionally dependent on the primary key.

A university uses the following relation:

Student (IDSt, StudentName, IDProf, ProfessorName, Grade)

The attributes IDSt and IDProf are the identification keys.

All attributes a single valued (1NF).

The following functional dependencies exist:

1. The attribute ProfessorName is functionally dependent on attribute IDProf (IDProf --> ProfessorName)
2. The attribute StudentName is functionally dependent on IDSt (IDSt --> StudentName)
3. The attribute Grade is fully functional dependent on IDSt and IDProf (IDSt, IDProf --> Grade)

Students

IDSt	LastName	IDProf	Prof	Grade
1	Mueller	3	Schmid	5
2	Meier	2	Borner	4
3	Tobler	1	Bernasconi	6

Startsituation

Result after normalisation

Students		Professors	
ID	LastName	IDProf	Professor
1	Mueller	1	Bernasconi
2	Meier	2	Borner
3	Tobler	3	Schmid

Grades

IDStIDProf	Grade
1 3	5
2 2	4
3 1	6

Example Second normal form

The table in this example is in first normal form (1NF) since all attributes are single valued. But it is not yet in 2NF. If student 1 leaves university and the tuple is deleted, then we lose all



information about Professor Schmid, since this attribute is fully functional dependent on the primary key IDSt. To solve this problem, we must create a new table Professor with the attribute Professor (the name) and the key IDProf. The third table Grade is necessary for combining the two relations Student and Professor and to manage the grades. Besides the grade it contains only the two IDs of the student and the professor. If now a student is deleted, we do not lose the information about the professor.

Third normal form (3NF)

Third normal form:

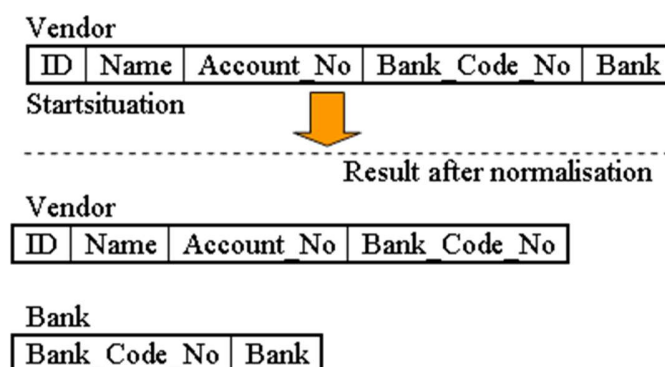
A relation is in third normal form if it is in 2NF and no non key attribute is transitively dependent on the primary key. A bank uses the following relation:

Vendor(ID, Name, Account_No, Bank_Code_No, Bank)

The attribute ID is the identification key. All attributes are single valued (1NF). The table is also in 2NF.

The following dependencies exist:

1. Name, Account_No, Bank_Code_No are functionally dependent on ID (ID --> Name, Account_No, Bank_Code_No)
2. Bank is functionally dependent on Bank_Code_No (Bank_Code_No --> Bank)



Example Third normal form

The table in this example is in 1NF and in 2NF. But there is a transitive dependency between Bank_Code_No and Bank because Bank_Code_No is not the primary key of this relation. To get to the third normal form (3NF), we have to put the bank name in a separate table together with the clearing number to identify it.



Exercise 1

The following table is already in first normal form (1NF). There is only one entry per field. Please convert this table to the third normal form (3NF) using the techniques you learned in this Unit. Write a short report about your solution and post it in the discussion board. Check the other solutions and comment them, if necessary. If you have questions, you can also post them in the discussion board. A tutor will look at the questions regularly and give feedback and answers.

A table with the students and their grades in different topics.

UnitID	StudentID	Date	TutorID	Topic	Room	Grade	Book	TutEmail
U1	St1	23.02.03	Tut1	GMT	629	4.7	Deumlich	tut1@fhbb.ch
U2	St1	18.11.02	Tut3	Gln	631	5.1	Zehnder	tut3@fhbb.ch
U1	St4	23.02.03	Tut1	GMT	629	4.3	Deumlich	tut1@fhbb.ch
U5	St2	05.05.03	Tut3	PhF	632	4.9	Dümmlers	tut3@fhbb.ch
U4	St2	04.07.03	Tut5	AVQ	621	5.0	SwissTopo	tut5@fhbb.ch

Exercise 2

The following table with name BUYERS_HOUSES is used by a construction company which is active in the construction of houses in a small city. In the table BUYERS_HOUSES data is stored for the houses that has been sold by the construction company to its customers. The construction company has divided the city into zones. Every home that is built and sold belongs to a zone. The selling price of one square meter is fixed within a zone.

**BUYERS_HOUSES**

HID	Zone	Price	ASM	Buyer Tax ID	Buyer last Name	Buyer First Name
123	A	1700	50	123	Adam	George
124	A	1700	75	234	Billy	Maria
125	B	2000	120	345	O' Sallivan	Anton
126	B	2000	110	456	Black	John
127	A	1700	90	567	Kindman	Bill
128	C	2400	70	678 901	Opaldon Ellonger	Geffrey Ann
129	A	1700	50	123	Adam	George

Where:

HID: the code given by the construction company in each house that manufactures. It is a unique number.

Zone: the zone to which a house belongs.

Price: the selling price per square metre of a house.

ASM: the area in square meters of a house

Buyer Tax ID: the tax registration number of a buyer of a home.

Buyer last Name: the surname of the buyer of a house.

Buyer First Name: the name of the buyer of a home.

a) Give an example of each of the anomalies observed in the BUYERS_HOUSES table.

b) Normalize the BUYERS_HOUSES table up to the 3rd normal form.



Exercise 3: 1st Normal Form (1NF)

Consider the Students table, with the primary key underlined, and the following data:

Students:

<u>Alpha</u>	Name	Email	Courses	GradePoints
100111	John Doe	doe@usna.edu	NN204, SI204, IT221	2,3,3
092244	Matt Smith	smith@usna.edu	SM223, EE301	4,4
113221	Melinda Black	black@usna.edu	SI204	3
090112	Tom Johnson	Johnson@usna.edu	NN204, SI204, IT221	4,2,3

- Is the Students table in 1NF? Why?
- If the Students table is not in 1NF, redesign the tables such that all the information currently in the Students table is found in the resulting tables, and the resulting tables are in 1NF. For each of the resulting tables, give the table name, column names, primary keys, and foreign keys.

Exercise 4: 3rd Normal Form (3NF)

For a table to be in 3rd form, the table must be in 1NF and 2nd NF. Below is the Rentals table created for the DVD-by-mail division of Neatflix.

Rentals

<u>RentalID</u>	<u>Title</u>	CustomerID	MailedOutDate	Director	MovieCategory	Price
1	Die Hard	1001	3/3/2010	John McTiernan	Old	\$4.25
1	The last man standing	1001	3/3/2010	Walter Hill	Old	\$4.25
1	Wedding Crashers	1001	3/3/2010	David Dobkin	New	\$5.50
2	Dodgeball	1002	3/4/2010	Rawson Marshall Thurber	New	\$5.50
2	Die Hard	1002	3/4/2010	John McTiernan	Old	\$4.25
3	As good as it gets	1003	1/7/2011	James Brooks	Old	\$4.25
4	Forest Gump	1001	1/7/2011	Robert Zemeckis	Old	\$4.25



The primary key of the Rentals table is the composite key (**RentalID, Title**).

- a) Explain the conditions under which the following functional dependency is true:
RentalID --> CustomerID
- b) Based on the sample data on the table, is the functional dependency
RentalID --> CustomerID true?
- c) Explain the conditions under which the following functional dependency is true:
Director --> Title
- d) Based on the sample data on the table, is the functional dependency
Director --> Title true?
- e) Based on your general knowledge of movies and rentals, is the functional dependency
Director --> Title true?
- f) Write a functional dependency that expresses the fact that the cost of all movies in a given category is the same.
- g) We discussed insertion anomalies, deletion anomalies and update anomalies as examples of problems that can appear in tables that are not normalized. The following is an example of an insertion anomaly in the Rentals table: if we want to create a new category of movies, "Must See", there is no way to store the price of this type of movie in the database, until someone rents a movie in this category, and the rental information is recorded into the Rentals table. Give one example of a deletion anomaly in the Rentals table.
- h) State what you believe are reasonable functional dependencies for the Rentals table for a DVD-by-mail business (include the functional dependencies from points a) to f) that you believe are/should be true).
- i) Given your answer above, decompose the Rentals table such that the resulting tables are in 3rdNF. For each of the resulting tables, give the table name, column names, primary keys, and foreign keys.

References



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- **ELMASRI, R., NAVATHE, S.B.**, 1994. *Fundamentals of Database Systems* 2nd. Redwood City, California: Addison-Wesley. Introduction in databases and SQL (in English).
 - **ZEHNDER, C.A.**, 1998. *Informationssysteme und Datenbanken*. Zürich: vdf Hochschulverlag AG.